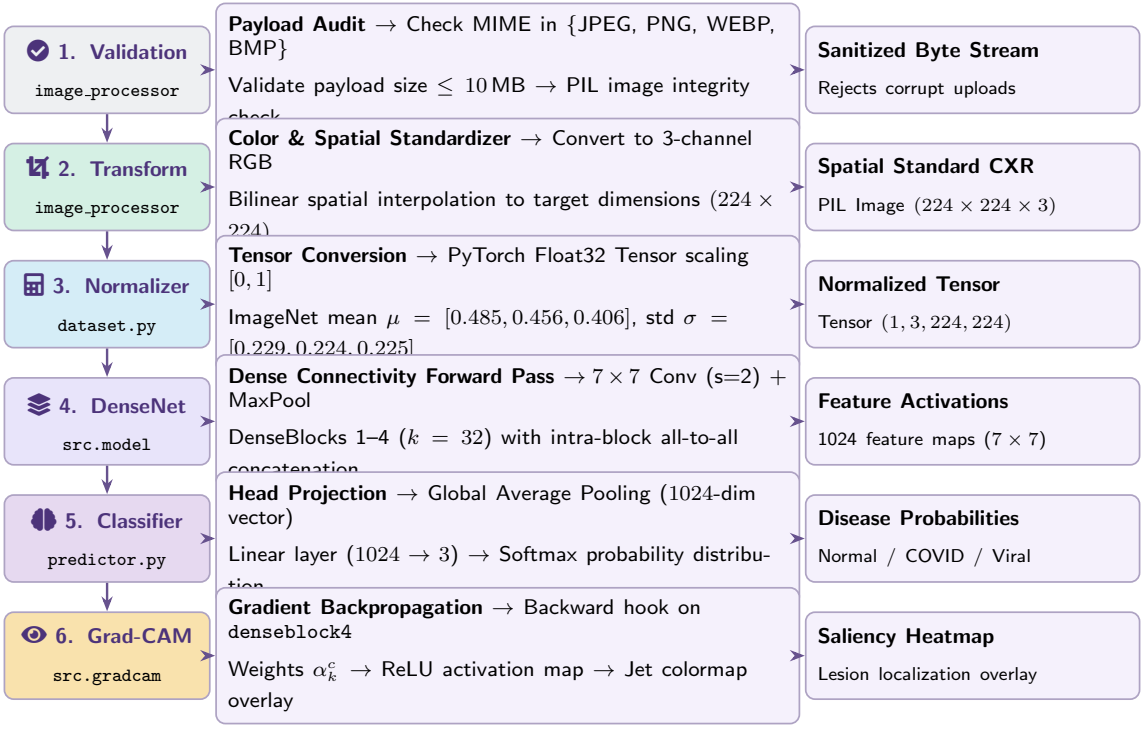

1. Validation
image_processor
Payload Audit → Check MIME in {JPEG, PNG, WEBP, BMP}
Validate payload size ≤ 10 MB → PIL image integrity check
Sanitized Byte Stream
Rejects corrupt uploads
2. Transform
image_processor
Color & Spatial Standardizer → Convert to 3-channel RGB
Bilinear spatial interpolation to target dimensions (224 × 224)
Spatial Standard CXR
PIL Image (224 × 224 × 3)
3. Normalizer
dataset.py
Tensor Conversion → PyTorch Float32 Tensor scaling [0, 1]
ImageNet mean μ = [0.485, 0.456, 0.406], std σ = [0.229, 0.224, 0.225]
Normalized Tensor
Tensor (1, 3, 224, 224)
4. DenseNet
src.model
Dense Connectivity Forward Pass → 7 × 7 Conv (s=2) + MaxPool
DenseBlocks 1–4 (k = 32) with intra-block all-to-all concatenation
Feature Activations
1024 feature maps (7 × 7)
5. Classifier
predictor.py
Head Projection → Global Average Pooling (1024-dim vector)
Linear layer (1024 → 3) → Softmax probability distribution
Disease Probabilities
Normal / COVID / Viral
6. Grad-CAM
src.gradcam
Gradient Backpropagation → Backward hook on denseblock4
Weights α_k^c → ReLU activation map → Jet colormap overlay
Saliency Heatmap
Lesion localization overlay